

Jiwon Kang

Last update: August 13, 2026

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Research Interests

Having started my research career in 3D vision, I expanded my focus during my M.S. to Generative AI and Unified Multimodal Models (UMMs), leading to four first-author papers at NeurIPS, ICLR, CVPR, and ECCV. Building on these experiences, I am now interested in world modeling that combines 3D visual capabilities with generative knowledge, with potential applications to real-world robotics and immersive gaming environments. Key research areas include:

- Diffusion & Multimodal Generative AI (Image/Video/Action/Audio Generation, Guidance, Sampling)
- Unified Multimodal Models
- 3D Vision (Novel View Synthesis, NeRF, Gaussian Splatting)

Education

M.S. in Artificial Intelligence

Kim Jaechul Graduate School of AI, KAIST

Mar. 2025 – Feb. 2027 (Expected)

Advisor: [Seungryong Kim](#) (CVLAB)

B.S. in Computer Science and Engineering

Korea University

Mar. 2019 – Feb. 2025

- GPA: 4.24 / 4.5
- Double Major in Statistics
- Leave of Absence for R.O.K. Military Service (Dec. 2020 – Jun. 2022)
- Undergraduate Research Intern @ [CVLAB](#) (Sep. 2022 – Feb. 2025), Advisor: Seungryong Kim

Experience

Kakao

Research Intern

Mar. 2026 – May 2026

Seoul, Korea

- Worked on Text-to-Speech research, repurposing Whisper as a continuous tokenizer for autoregressive diffusion-based TTS model [\[P2\]](#).

Projects

SAMSUNG

Research Collaboration

Aug. 2025 – Oct. 2025

Seoul, Korea

- Collaborated with Samsung Electronics on background face anonymization via generative face-swapping [\[C3\]](#).

Publications

*: Equal Contribution, †: Corresponding Author

International Conference

[C4] Transferability Between Understanding and Generation in Unified Multimodal Models [Paper](#) | [Project Page](#) | [Code](#)

Jiwon Kang^{*}, Heeji Yoon^{*}, Jaewoo Jung, Jaewon Min, Minkyong Jeon, Biyeon Hwang, Sangwon Jung, Seungryong Kim[†]

- Investigated cross-task transferability between image understanding and generation in Unified Multimodal Models, revealing that its strength strongly depends on architecture and emerges most clearly with a fully shared Transformer backbone and unified visual encoder.
- Leveraged understanding-to-generation transfer as a practical post-training strategy to improve targeted generative capabilities, including counting, spatial relations, and text rendering, while mitigating the distribution shift and quality degradation caused by direct generation fine-tuning.

European Conference on Computer Vision (ECCV 2026).

[C3] APPLE: Attribute-Preserving Pseudo-Labeling for Diffusion-Based Face Swapping [Paper](#) | [Project Page](#) | [Code](#)

Jiwon Kang^{*}, Yeji Choi^{*}, JoungBin Lee, Wooseok Jang, Jinhyeok Choi, Taekeun Kang, Yongjae Park, Myungin Kim, Seungryong Kim[†]

- Identified attribute loss in diffusion-based face swapping caused by masked inpainting conditions, and designed an attribute-preserving teacher using conditional deblurring and attribute-aware inversion to retain both global and fine-grained target attributes.
- Constructed high-fidelity pseudo-triplets from the teacher to train a direct-editing student on clean attribute conditions, achieving state-of-the-art attribute preservation with competitive identity transfer and consistently lying on or near the Pareto frontier across datasets.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026).

[C2] A Noise is Worth Diffusion Guidance [Paper](#) | [Project Page](#) | [Code](#)

Donghoon Ahn^{*}, **Jiwon Kang**^{*}, Sanghyun Lee, Jaewon Min, Wooseok Jang, Minjae Kim, Hyungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha[†], Kyong Hwan Jin[†], Seungryong Kim[†]

- Revealed that high-quality guidance-free diffusion is largely determined by the initial noise, identifying small low-magnitude, low-frequency components that transform Gaussian noise into a favorable noise space for unguided denoising.
- Proposed NoiseRefine, which learns to refine the initial noise once before sampling, reproducing guidance-level generation quality without additional guidance passes while reducing inference cost and requiring only 50K text-image pairs for training.

International Conference on Learning Representations (ICLR 2026).

[C1] Where and How to Perturb: On the Design of Perturbation Guidance in Diffusion and Flow Models [Paper](#) | [Project Page](#) | [Code](#)

Donghoon Ahn^{*}, **Jiwon Kang**^{*}, Sanghyun Lee, Minjae Kim, Wooseok Jang, Jaewon Min, Sangwu Lee, Sayak Paul, Seungryong Kim[†]

- Conducted the first head-level analysis of attention perturbation guidance in Diffusion Transformers, revealing that individual attention heads exhibit specialized and compositional effects on visual concepts such as structure, lighting, style, and texture.
- Proposed HeadHunter to iteratively select attention heads for arbitrary user-defined objectives and SoftPAG to continuously control perturbation strength, enabling fine-grained quality and style guidance on SD3 and FLUX.1 with compact head subsets and negligible inference overhead.

Conference on Neural Information Processing Systems (NeurIPS 2025).

Workshop

[W1] Self-Evolving Neural Radiance Fields [Paper](#) | [Project Page](#) | [Code](#)

Jaewoo Jung^{*}, Jisang Han^{*}, **Jiwon Kang**^{*}, Seongchan Kim, Min-Seop Kwak, Seungryong Kim[†]

- Formulated few-shot NeRF as an iterative teacher-student self-training problem, progressively expanding supervision with pseudo-labeled novel views generated by the teacher and promoting each trained student to the next teacher.
- Introduced cross-view semantic consistency based ray reliability estimation and reliability-aware distillation, allowing existing NeRF models to learn more robust geometry from sparse views without external 3D priors or architectural modifications.

ICCV 2025 Workshop on Wild3D: 3D Modeling, Reconstruction, and Generation in the Wild.

Preprints & Under Review

[P2] Exploring Whisper as a Continuous Tokenizer for Text-to-Speech

Jiwon Kang, Heeji Yoon, Woo-Young Kang, Sunghun Kang, Jihwan Eom, Jaeyeol Jeon, Byungseok Roh, Seungryong Kim[†]

- Inspired by the success of RAE in the vision domain, explored whether ASR-supervised audio representations (Whisper) could be repurposed as generative representations for TTS.
- Developed (1) a Whisper-based continuous tokenizer and (2) a autoregressive diffusion-based TTS model that generates Whisper tokens, enabling high-quality speech synthesis with controllable prosody.

Under Review.

[P1] RAIN-GS: Relaxing Accurate Initialization Constraint for 3D Gaussian Splatting [Paper](#) | [Project Page](#) | [Code](#)

Jaewoo Jung^{*}, Jisang Han^{*}, Honggyu An^{*}, Jiwon Kang^{*}, Seonghoon Park^{*}, Seungryong Kim[†]

- Analyzed why 3D Gaussian Splatting critically depends on accurate SfM initialization, showing that limited Gaussian transport and missing low-frequency scene structure prevent robust coarse-to-fine optimization from random or noisy point clouds.
- Proposed RAIN-GS with Sparse-Large-Variance initialization, progressive Gaussian low-pass filtering, and Adaptive Bound-Expanding Split, enabling 3DGS to train from sub-optimal or random point clouds with performance comparable to or better than SfM initialization.

Under Review.

Honors

Outstanding Paper Silver Award, IPIU 2024

2024

- Paper: *Self-Evolving Neural Radiance Fields*

Extracurricular Activities

Military Service (KATUSA, Korean Augmentation to the U.S. Army)

Dec. 2020 – Jun. 2022

- MOS : 92Y (Unit Supply Specialist)
- 194th DSSB, 2nd Infantry Division, Eighth U.S. Army, Camp Humphreys

AIKU (Artificial Intelligence Society of Korea University)

Sep. 2022 – Aug. 2023

- Co-founder & Vice President of academic research society for undergraduate students